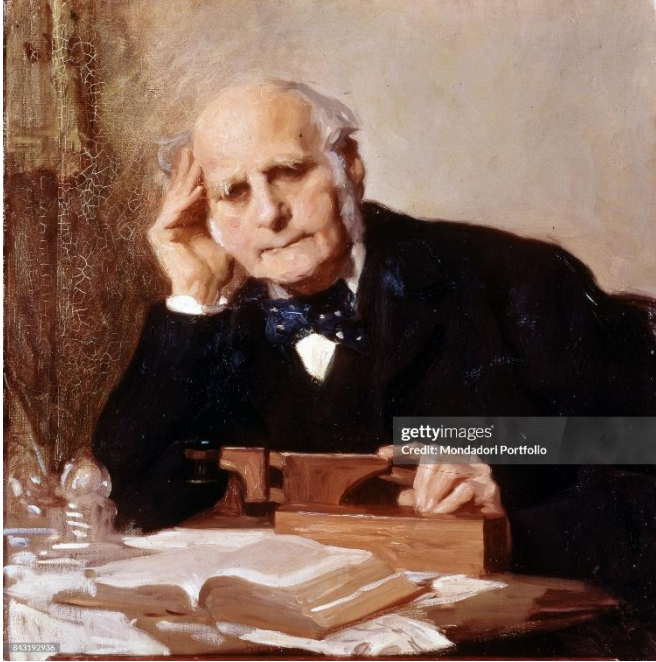


Multivariate Data Analysis and Process Modeling

- a) Linear Regression
- b) Essential Linear Algebra
- c) Logistic Regression
- d) Basic Multivariate data calculations and statistical distance
- e) Principal Component Analysis
- f) Neural Networks
- g) Discriminant Analysis
- h) For Design of Experiments pl. see link <https://nptel.ac.in/courses/103106112>

Regression Analysis



“It appeared from these experiments that the offspring did not tend to resemble their parents in size, but always to be more mediocre than they – to be smaller than the parents, if the parents were large; to be larger than the parents, if the parents were small.”

Sir Francis Galton

Prescribed Text Books

Montgomery, D. C., G.C. **Runger**, *Applied Statistics and Probability for Engineers*. 5th ed. New Delhi: Wiley-India, 2011.

Draper, N. R., H. Smith, *Applied Regression Analysis*. 3rd ed. New Delhi: Wiley-India, 1998.

Kutner, M. H., C. J. Nachtschiem, J. Netner, *Applied Linear Regression Models*. 4th ed. New Delhi: McGraw Hill, 2004.

Note: *In case of any reference be inadvertently omitted, the error will be rectified if intimated*

What is Meant by Regression?

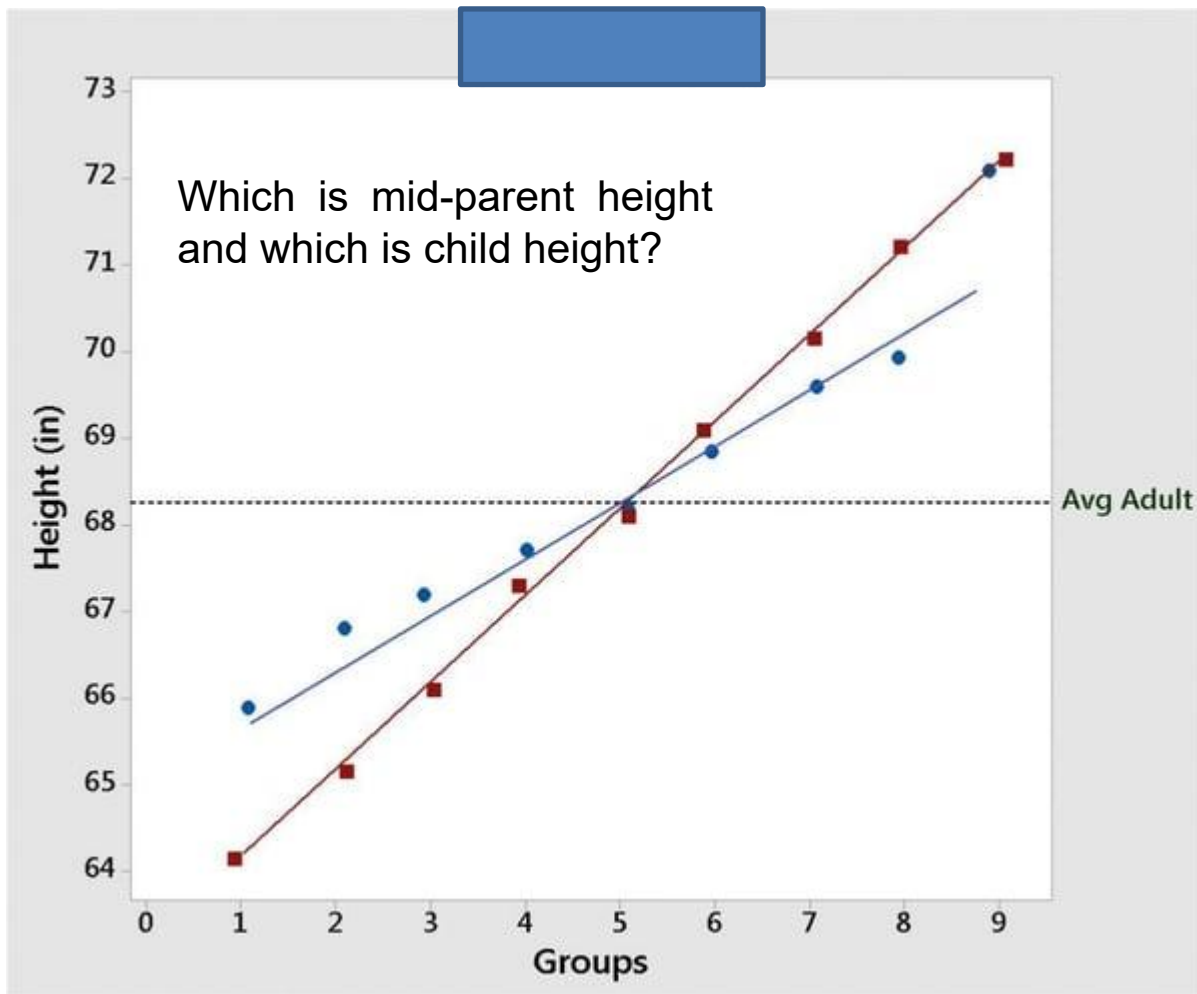
- ❖ A return to former state of existence
- ❖ Reduced symptoms or lower severity of a disease instead of its progression (regression of disease)
- ❖ The relation between input and output is called regression because it refers to a statistical tendency of data points to move ("regress" or "step back") towards the average (mean) value.

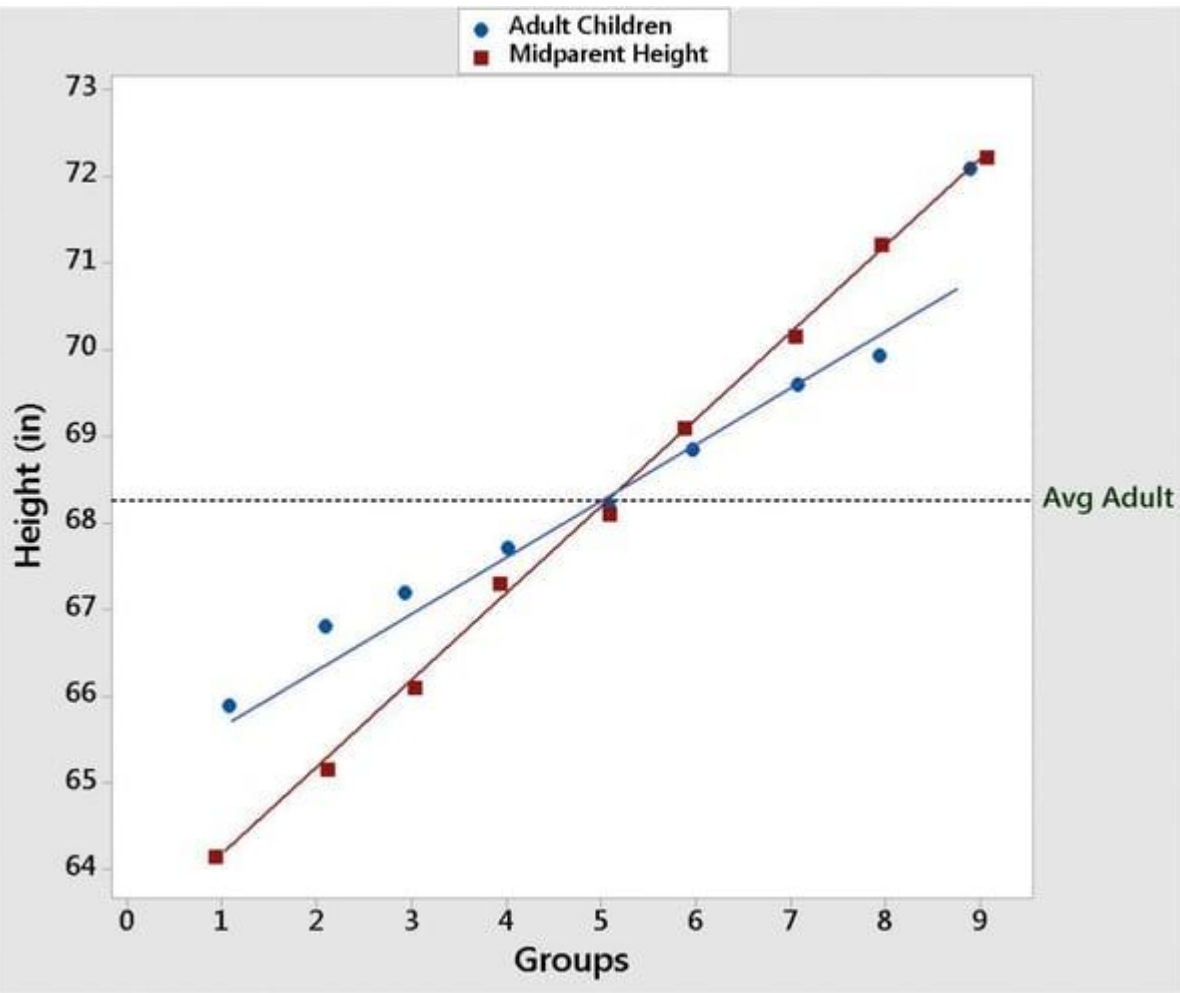
What is Meant by Regression?

- ❖ Historical Origin: "Regression to the Mean"
- ❖ Galton's Observations (1870s-1880s): While studying the inheritance of traits, Galton observed that children of very tall parents were usually shorter than their parents but taller than the average, while children of very short parents were usually taller than their parents but shorter than the average.

What is Meant by Regression?

- ❖ **Regressing Toward Mediocrity:** Galton called this phenomenon "regression toward mediocrity" or "regression to the mean," meaning that data points extreme in value tend to "regress" back toward the average value.
- ❖ **The "Line of Regression":** To visualize and model this, Galton used a straight line to summarize the relationship, which he termed the "regression line"
- ❖ **Machine Learning Definition:** In modern data science, regression refers to a supervised learning technique used to predict a **continuous numerical output** (e.g., price, temperature, sales volume) based on input features.





The Regression Line of Galton

- ❖ Regression to the mean is an important statistical phenomenon which is often neglected and may lead to misleading conclusions.
- ❖ For example, statistics on the impact of CCTVs indicated that they reduced crime by around 100 robberies
- ❖ This inference was based on the reduction in crime after the CCTVs had been installed.
- ❖ But, CCTVs are often installed after there have been an unusually high number of robberies
- ❖ So you may generally anticipate that the number of robberies return to normal levels afterwards anyway

Regression and Causality

- ❖ Let there be a strong statistical relation between the response variable Y and the explanatory or predictor variable X
- ❖ But it does not imply in any way that Y depends causally on X i.e., X was solely responsible for causing Y
- ❖ Irrespective of the strong statistical relation between X and Y , do not attribute a cause-and-effect pattern to the regression model.

What is Linear about Regression?

- ❖ In linear regression, "linear" means the model is linear in its parameters ($\beta_0, \beta_1, \dots$)
- ❖ This means that the parameters are not multiplied, divided, or raised to powers, allowing the model to be expressed as a weighted sum:

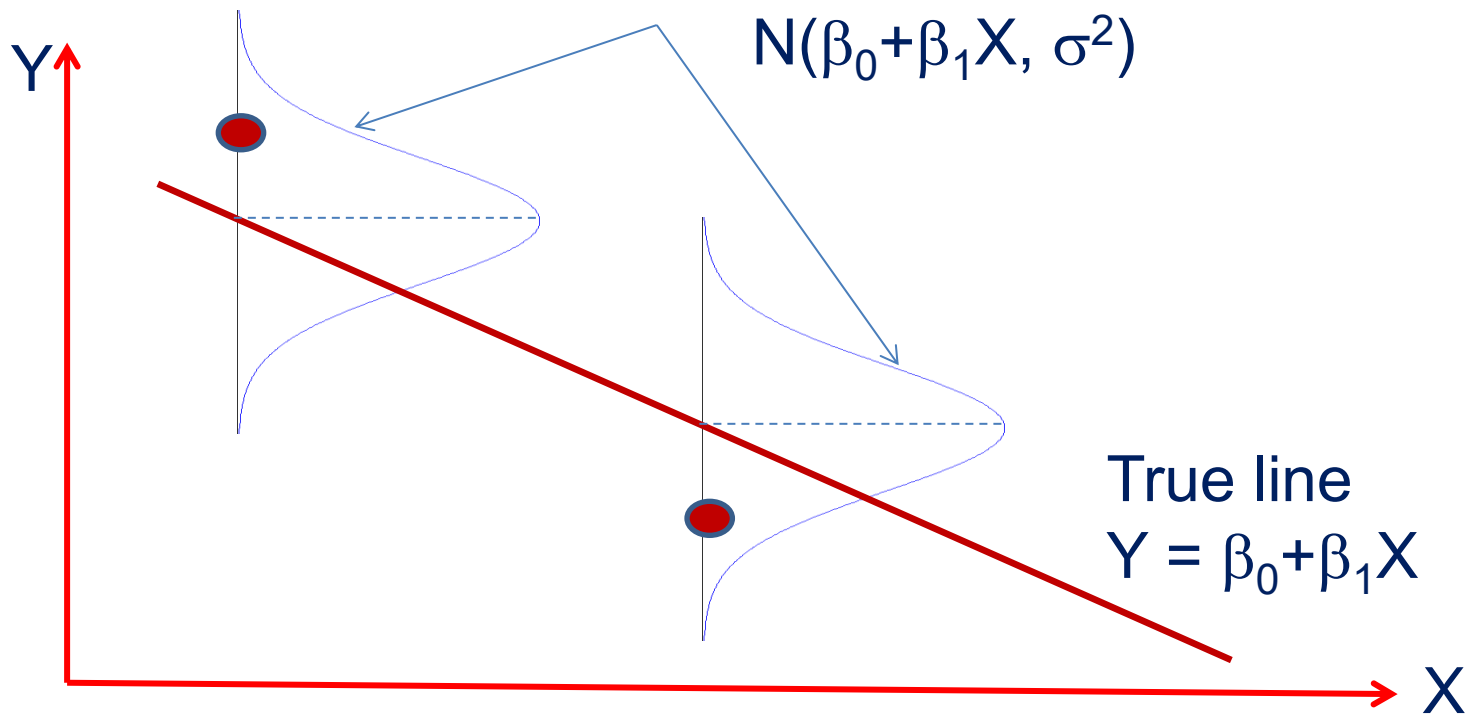
$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

What is Linear about Regression?

Example Equations:

- **Linear:** $y = \beta_0 + \beta_1 x$ (Linear in parameters and variables) $+ \varepsilon$
- **Linear:** $y = \beta_0 + \beta_1 x + \beta_2 x^2$ (Linear in parameters, non-linear in variable x) $+ \varepsilon$
- **Non-linear:** $y = \beta_0 + \beta_1^2 x$ (Not linear, because parameter β_1 is squared) $+ \varepsilon$

Nonlinear Regression: $Y = \beta_0 + e^{\beta_1 X} + \epsilon$. Here, β_1 is an exponent; it is **nonlinear in parameters**. You cannot solve this using standard linear algebra (Ordinary Least Squares) because the parameter is "trapped" inside a nonlinear function.



Overall Summary of Regression

- ❖ Regression model assumes that the probability distributions of Y have the same variance σ^2 , regardless of the level of the predictor variable X .
- ❖ The error terms are assumed to be uncorrelated.
- ❖ Since the error terms ε_i and ε_j are uncorrelated, so are the responses Y_i and Y_j
- ❖ Regression model implies that the responses Y_i come from probability distributions whose means are $E(Y_i) = \beta_0 + \beta_1 X_i$ and whose variances are σ^2 , the same for all levels of X

Underlying Assumptions in Linear Regression

- ❖ **A1:** Linearity in parameters
- ❖ **A2:** The independent variables are measured without error
- ❖ **A3: Homoscedasticity:** Error terms have a constant variance: $\text{Var}(\varepsilon_i) = \sigma^2$
- ❖ **A5:** The error terms ε_i are normally distributed: $\varepsilon_i \sim N(0, \sigma^2)$
- ❖ **A6:** No perfect multi-collinearity

Another Important Assumption

$$\mathbf{A0: Cov}(\varepsilon_i, X_{ji}) = 0$$

In linear regression, this means that the error term (ε_i) is **uncorrelated** with the i^{th} setting of the independent variables (X_j).

This fundamental assumption, often called **exogeneity**, ensures that the independent variables do not explain the variance in the errors

Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

Y : Response

X_1 : independent variable 1

X_2 : independent variable 2

ε : independent error term

Multiple Linear Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

The term **multiple** appears because it contains more than one regressor variable and the term linear appears because the equation is a linear function of the unknown parameters β_0 , β_1 and β_2 .

Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon$$

$$5 + 0.2 \overset{0(0)}{X_1} + 50 \overset{(0.1)}{X_2}$$

$$5 + 50 X_2$$

β_0 : Intercept

β_1 : partial regression coefficient 1

β_2 : partial regression coefficient 2

$$\frac{X - \mu}{\text{Std dev.}}$$

Normalize data

X_1	X_2	Y
1	1000	9
2	5000	16
3	7000	23
<u>32</u>	<u>84.33×10^3</u>	
	$5 + 50 X_2^*$	$1 + 50 X_2^*$

Partial Regression Coefficients

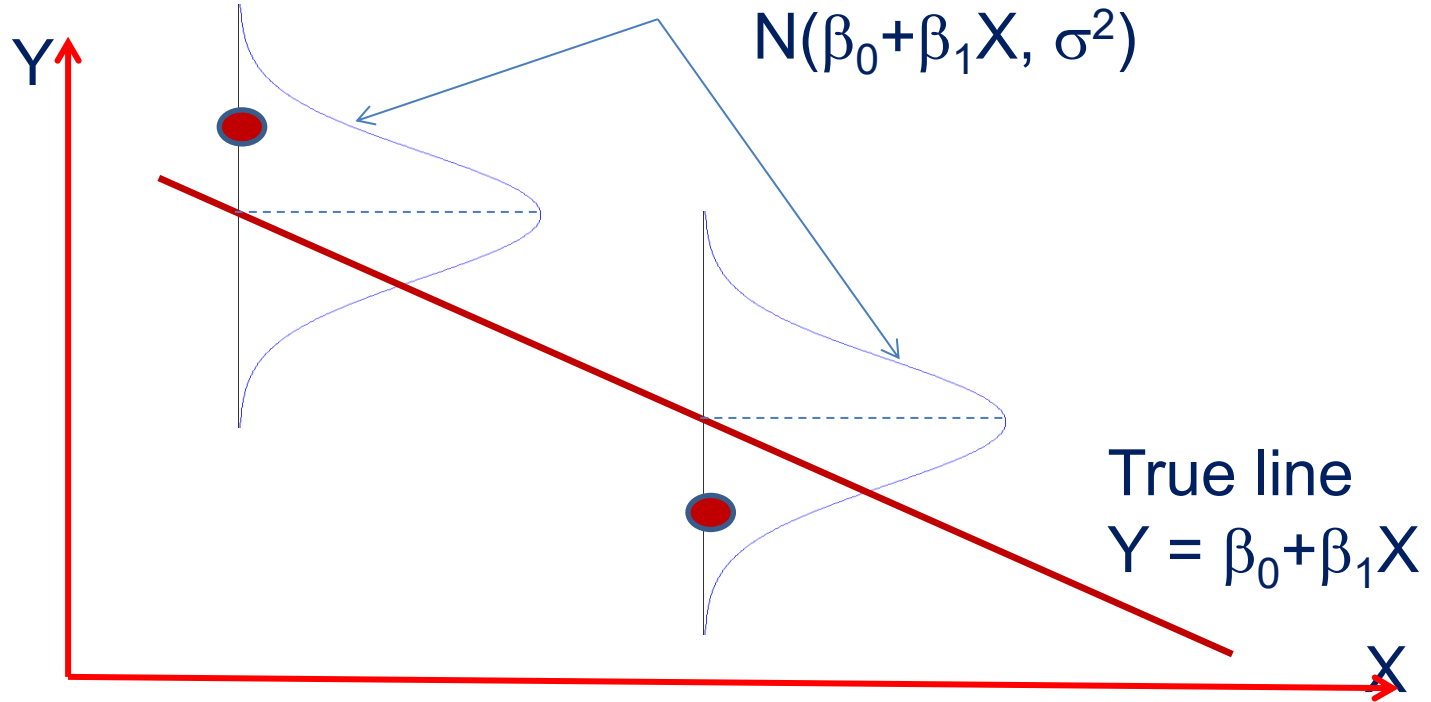
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon \quad \text{model}$$

β_1 and β_2 : **partial regression coefficients** 1 and 2

The term **partial** is used because

β_1 refers to the change in the response Y due to a change only in X_1 with X_2 being kept constant.

Regression: Illustration



Partial Regression Coefficients

- ❖ Each response is assumed to belong to a normal distribution centered vertically at the coordinate given by the regression line.
- ❖ The variances of all the normal distributions are assumed to be the same.

Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

This is a multiple regression model with **k regressor variables ($X_1, X_2, \dots, X_k$)**.

The parameters $\beta_0, \beta_1, \beta_2, \dots, \beta_k$ are called as partial regression coefficients.

Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_k X_k + \varepsilon$$

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

~~+ ε~~

$0-20$
Do not use · 30
 $30-2000$
10
 $-50+5$
- 10

Even though the actual functional relationship between Y and $X_1, X_2, \dots, X_k$ is unknown, the regression equation represents an *approximate relationship* between the two over a *certain (narrow) range* of the variables.

Multiple Regression Model

The Multiple Regression Model involving **interaction** between variables may be represented by

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{12} \overbrace{X_1 X_2}^{X_3} + \varepsilon$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_1^2$$

$$+ \beta_5 X_2^2 + \varepsilon$$

$$\begin{bmatrix} X_1^2 & X_1 & X_2 & X_1 X_2 & X_1^2 \\ 4 & 1 & 2 & 2 & 1 \\ 1 & -4 & 1 & -4 & 16 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \vdots & \vdots \end{bmatrix}$$

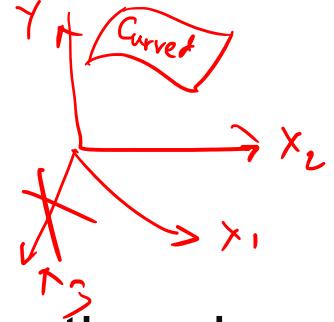
Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{12} X_1 X_2 + \varepsilon$$

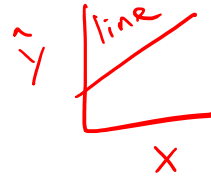
This is still a linear regression model as the functional relationship is linear in the unknown parameters (β_0 , β_1 , β_2 , β_{12}).

Multiple Regression Model

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{12} X_1 X_2 + \varepsilon$$

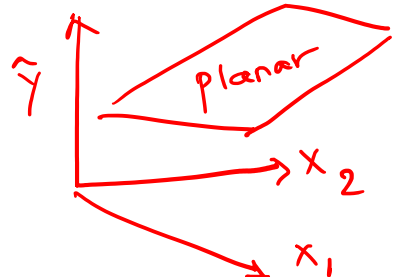


Expressing $X_1 X_2$ as equal to X_3 we may write the above equation as



$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_1$$
$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \hat{\beta}_2 x_2$$

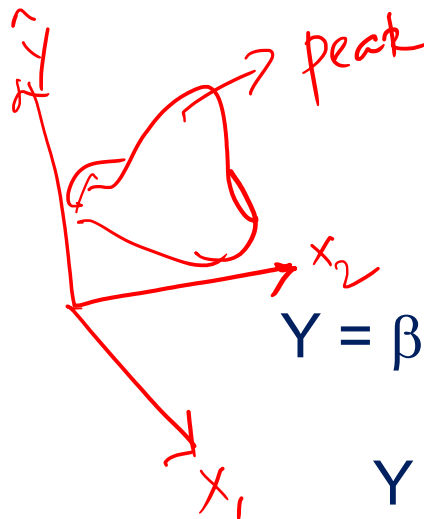
$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \varepsilon$$

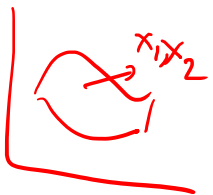


Multiple Regression Model

- ❖ When we generate a response surface to this model, we get a curved surface.
- ❖ However this still remains a linear model as it is linear in its parameters ($\beta_0, \beta_1, \beta_2, \beta_3$)
- ❖ Without interaction term, we have a flat planar surface

Multiple Regression Model


$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_{11} X_1^2 + \beta_{22} X_2^2 + \beta_{12} X_1 X_2 + \varepsilon$$

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4 + \beta_5 X_5 + \varepsilon$$


Any regression model that is linear in parameters (β s) is a linear regression model, irrespective of the resulting response surface being a plane or curved.

Matrix Approach to Multiple Regression

Let there be k regressor variables with n observations

$$(X_{i1}, X_{i2}, X_{i3}, \dots, X_{ik}, Y_i), i=1, 2, \dots, n$$

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_k X_{ik} + \varepsilon_i \quad i=1, 2, \dots, n$$

$$\begin{matrix} & X_1 & X_2 & Y \\ i=1 & X_{11} & X_{12} & Y_1 \\ 2 & X_{21} & X_{22} & Y_2 \\ \vdots & \vdots & \vdots & \vdots \\ n & X_{n1} & X_{n2} & Y_n \end{matrix}$$

Note: $n > k$

$$\begin{matrix} \text{column} & j & k \text{ regressors} & X_1 & X_2 & \dots & X_k \\ \text{row} & i & n \text{ runs} & 1 & 2 & \dots & n \end{matrix}$$

This may be represented in a matrix notation as $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$

Matrix Form of the Regression Equations

$$Y_1 = \beta_0 + \beta_1 X_{11} + \beta_2 X_{12} + \dots + \beta_k X_{1k}$$

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_i \\ \vdots \\ Y_n \end{bmatrix}$$

$$X = \begin{bmatrix} 1 & X_{11} & X_{12} & X_{13} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & X_{23} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & X_{n1} & X_{n2} & X_{n3} & \dots & X_{nk} \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix}$$

$$\epsilon = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$



$$Y = X\beta + \epsilon$$

$$k=3$$

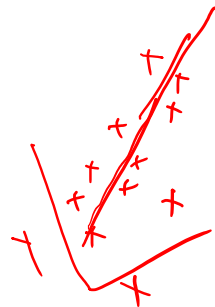
$$p=3+1$$

$$\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \hat{\beta}_3$$

$$= X_1, X_2, X_3$$

partial regression coefficients

4 unknowns
4 eqns.



$$\mathbf{Y} = \begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_n \end{bmatrix}$$

$n \times 1$

$$\mathbf{X} = \begin{bmatrix} 1 & X_{11} & X_{12} & X_{13} & \dots & X_{1k} \\ 1 & X_{21} & X_{22} & X_{23} & \dots & X_{2k} \\ \vdots & \vdots & \vdots & \vdots & \dots & \vdots \\ 1 & X_{n1} & X_{n2} & X_{n3} & \dots & X_{nk} \end{bmatrix}$$

$$n \times \underbrace{(1+k)}_p = n \times p$$

$\mathbf{Y}$ is an $(n \times 1)$ vector of the observations

$\mathbf{X}$ is an $(n \times p)$ matrix of the levels of the independent variables

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \vdots \\ \vdots \\ \beta_k \end{bmatrix} \quad \begin{matrix} k \times 1 \\ (k+1) \times 1 \\ p \times 1 \end{matrix}$$

$$\varepsilon = \begin{bmatrix} \varepsilon_1 \\ \varepsilon_2 \\ \vdots \\ \vdots \\ \vdots \\ \varepsilon_n \end{bmatrix} \quad n \times 1$$

β is a $(p \times 1)$ vector of the regression coefficients, and ε is a $(n \times 1)$ vector of the random errors. Here $p = k+1$

Matrix Approach to Multiple Regression

$$Y = X\beta + \varepsilon$$

$n \times 1$ $n \times p$ $p \times 1$ $n \times 1$
 $n \times 1$ $n \times 1$ $n \times 1$

$$Y = \beta_1 x_1 + \beta_2 x_2$$

$$Y = \beta X$$

$p \times 1 \times n \times p$

We wish to find estimates for the linear regression model coefficients (also called parameters) which are expressed as follows $\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k$

Matrix Approach to Multiple Regression

$$\hat{\beta}_0, \hat{\beta}_1, \hat{\beta}_2, \dots, \hat{\beta}_k$$

Using these coefficients in the following model will help us to predict the response for various values of X_i ($i=1,2,\dots,k$)

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots + \hat{\beta}_k X_k$$

Least Squares Estimators of β

The least squares estimator $\hat{\beta}$ is the solution for β in the equations

$$\frac{\partial L}{\partial \beta} = 0$$

Where

$$L = \sum_{i=1}^n \varepsilon_i^2 = \boldsymbol{\varepsilon}'\boldsymbol{\varepsilon}$$

Least Squares Estimators of β

This leads to the solving of the following system of equations

$$\mathbf{X}'\mathbf{X}\hat{\beta} = \mathbf{X}'\mathbf{Y}$$

These are the **least squares model equations** in matrix form. Their solution is given by

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Dimensions of Matrices

$$\mathbf{X} : (n \times p)$$

$$\mathbf{X}'\mathbf{X} : (p \times n) \times (n \times p) = (p \times p)$$

$$\mathbf{X}'\mathbf{Y} : (p \times n) \times (n \times 1) = (p \times 1)$$

$$\hat{\boldsymbol{\beta}} : (p \times 1)$$

$$\mathbf{y} : (n \times 1)$$

Note that $p = k+1$

Dimensions of Matrices

The fitted regression model is given by $\hat{Y} = \mathbf{X}\hat{\beta}$

The difference between the actual i^{th} observation and the predicted value given above (called **residual**) may be

expressed in matrix notation as $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$

Webb
M Tech

kannan @gmail. iitm.ac.in
gmail → zebra

$$\begin{bmatrix} y_1 - \hat{y}_1 \\ y_2 - \hat{y}_2 \\ \vdots \\ y_n - \hat{y}_n \end{bmatrix}$$

$$= \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} - \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

Calculations of Error Sum of Squares (SS_E)

$e = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ $e = Y - \hat{Y}$ $e' = [1 \ 2]$ $e' = Y' - \hat{Y}'$

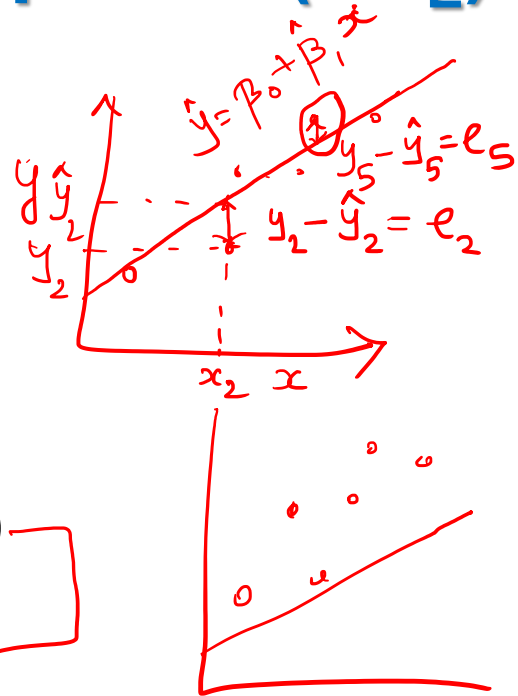
$\hat{Y} = X\hat{\beta}$

$e_2 + e_5$
 $e_2 + e_5$
 $e_2 + e_5$

It may be shown that

$$e'e = SS_E = (Y - X\hat{\beta})' (Y - X\hat{\beta})$$

$$SS_E = (Y'Y - \hat{\beta}'X'Y)$$



Variance-Covariance Matrix

The variances of the least square estimators are the elements of the $(\mathbf{X}'\mathbf{X})^{-1}$ matrix multiplied by the variance σ^2 .

$(\mathbf{X}'\mathbf{X})^{-1} \sigma^2$ is termed as the

Variance-Covariance matrix.

Variance-Covariance Matrix

The diagonal elements of the Covariance matrix are the **variances of the least square estimators** and the off-diagonal elements are the **covariances**.

The Covariance matrix dimensions are $p \times p$

Matrix Form of the Regression Equations

$$\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1}\sigma^2 = \begin{bmatrix} C_{00} & C_{01} & C_{02} \\ C_{10} & C_{11} & C_{12} \\ C_{20} & C_{21} & C_{22} \end{bmatrix} \sigma^2$$

Matrix Form of the Regression Equations

This is a symmetric matrix

($C_{10}=C_{01}$, $C_{20}=C_{02}$, and $C_{21}=C_{12}$) as $(\mathbf{X}'\mathbf{X})^{-1}$ is symmetric.

$$V(\hat{\beta}) = \sigma^2 C_{jj}, j = 0, 1, 2$$

$$\mathbf{cov}(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2 C_{ij}, i \neq j$$

Significance of the Covariance Matrix

Estimates of the variances of the regression coefficients may be obtained by substituting σ^2 with an estimate as σ^2 is unknown.

Estimated Standard Error

When σ^2 is substituted by its estimate $\hat{\sigma}^2$, the square root of the estimated variance of the j^{th} regression coefficient is termed as the **estimated standard error** of the least squares estimator $\hat{\beta}_j$

Estimated Standard Error

The **estimated standard error** of the least squares estimator is given by

$$se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$$

Estimated Standard Errors

- ❖ These standard errors provide an idea of the precision of the estimation for the regression coefficients
- ❖ Small standard errors indicate good precision.

Estimator of Error Variance

- ❖ Estimation of residual error is important
- ❖ The residual should ideally reflect the difference due to random factors and not systematic discrepancies created by using an inadequate model
- ❖ This is an unbiased estimator of σ^2

Estimator of Error Variance

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n e_i^2}{n - p} = \frac{SS_E}{n - p}$$

$$SS_E = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

Why n-p?

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^n e_i^2}{n - p} = \frac{SS_E}{n - p}$$

$$SS_E = \sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$$

$$SS_E = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- ❖ Two parameters are estimated here from the data.
- ❖ Two constraints operate on the sum of squares.
- ❖ The constraints operating on $\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$ are

$$\frac{\partial SS_E}{\partial \hat{\beta}_0} = \frac{\partial}{\partial \hat{\beta}_0} \left[\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \right] = 0$$

$$\frac{\partial SS_E}{\partial \hat{\beta}_1} = \frac{\partial}{\partial \hat{\beta}_1} \left[\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \right] = 0$$

Why n-p?

- ❖ Two parameters are estimated here from the data. Two constraints operate on the sum of squares.
- ❖ The constraints operating on $\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2$ are

$$\frac{\partial SS_E}{\partial \hat{\beta}_0} = \frac{\partial}{\partial \hat{\beta}_0} \left[\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \right] = 0$$

$$\frac{\partial SS_E}{\partial \hat{\beta}_1} = \frac{\partial}{\partial \hat{\beta}_1} \left[\sum_{i=1}^n (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i)^2 \right] = 0$$

Standard Error

$$\text{se}(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$$

- ❖ We compare the standard errors with the estimated coefficients and
- ❖ if the standard errors are much lower than the estimated coefficients, the **precision** of estimation is good.

Statistical Properties of the Coefficients

The least squares model equations in matrix form is given below. The solution is given by

$$\mathbf{X}'\mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{Y}$$

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

Actual response and its prediction

Express the deviation between an actual response and the predicted response in terms of difference of two deviations

- ❖ Deviation of actual response from mean response
- ❖ Deviation of predicted response from mean response

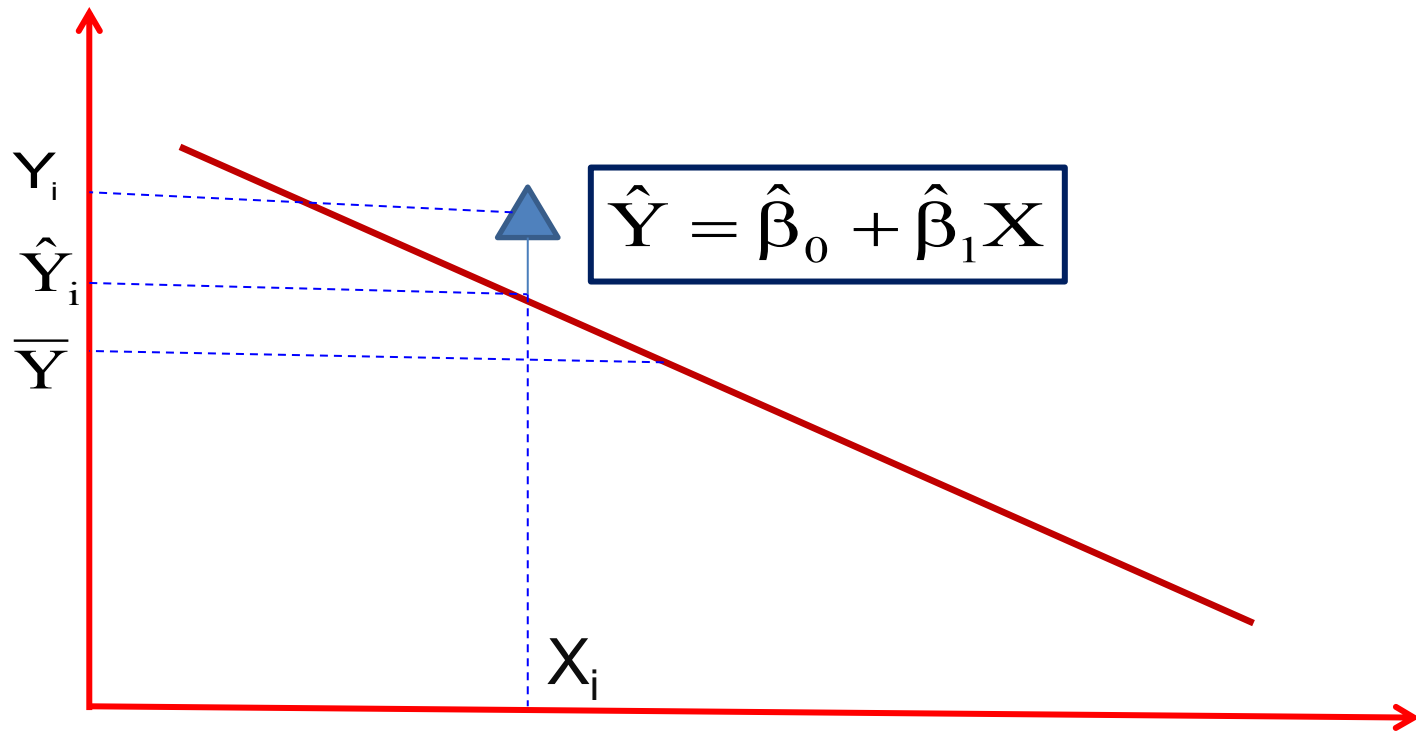
Relation between actual response and its prediction

- ❖ Deviation of actual response from mean response
- ❖ Deviation of predicted response from mean response

$$(Y_i - \bar{Y}) = (Y_i - \hat{Y}_i) + (\hat{Y}_i - \bar{Y})$$

or

$$(Y_i - \hat{Y}_i) = (Y_i - \bar{Y}) - (\hat{Y}_i - \bar{Y})$$



Sum of Squares in Linear Regression

Total Sum of Squares =

Regression Sum of Squares

+

Residual Sum of Squares

$$SS_{\text{Total}} = SS_{\text{Regression}} + SS_{\text{Residual}}$$

Resolution of Total Sum of Squares

The total sum of squares is given as usual by

$$SS_{\text{total}} = \sum_{i=1}^n (Y_i - \bar{Y})^2$$

Resolution of Total Sum of Squares

$$\begin{aligned}SS_{\text{total}} &= \sum_{i=1}^n (Y_i - \bar{Y})^2 \\&= \sum_{i=1}^n Y_i^2 - \frac{(\sum_{i=1}^n Y_i)^2}{n} \\&= \mathbf{Y}'\mathbf{Y} - \frac{(\sum_{i=1}^n Y_i)^2}{n}\end{aligned}$$

Resolution of Error Sum of Squares

$$SS_E = (Y'Y - \hat{\beta}'X'Y)$$

$$SS_E = (Y'Y - \frac{(\sum_{i=1}^n Y_i)^2}{n}) - (\hat{\beta}'X'Y - \frac{(\sum_{i=1}^n Y_i)^2}{n})$$

$$SS_{\text{error}} = SS_{\text{total}} - SS_{\text{regression}}$$

By definition of the total sum of squares, the second term becomes the **regression sum of squares**

Regression Sum of Squares due to β_0

Total Sum of Squares: $\mathbf{Y}'\mathbf{Y} - \frac{(\sum_{i=1}^n Y_i)^2}{n}$

The actual total sum of squares is $\mathbf{Y}'\mathbf{Y}$.

The term $\frac{(\sum_{i=1}^n Y_i)^2}{n}$ is deducted from the total sum of squares.

Regression Sum of Squares due to β_0

Total Sum of Squares: $Y'Y - \frac{(\sum_{i=1}^n Y_i)^2}{n}$

The regression sum of squares does not include the intercept β_0 contribution and has contributions only from

$$\beta_1, \beta_2, \beta_3, \dots, \beta_k$$

Regression Sum of Squares due to β_0

The contribution to the sum of squares due to β_0 is

$$\frac{(\sum_{i=1}^n Y_i)^2}{n} = n\bar{Y}^2$$

Regression Sum of Squares due to β_0

Another way of looking at this is as follows

If we fit a model containing only β_0 , the fitted model is

$$\hat{Y} = \bar{Y}$$

or

$$\beta_0 = \bar{Y}$$

Regression Sum of Squares due to β_0

When we do sum of squares analysis we first consider the contribution from the intercept (β_0) and then consider the *additional* contribution from the remaining regression coefficients for e.g. the slope (β_1) and so on.

Regression Analysis: A Few Terms Explained

Coefficient of Determination (R^2):

$$R^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}}$$

This represents the proportion of the total variability accounted or explained by the linear regression model

Coefficient of Determination (R^2):

The R^2 value may be increased by increasing the number of terms in the model and thereby increasing the number of coefficients that may be adjusted so that the model can be made to fit the data excellently.

Coefficient of Determination (R^2):

- ❖ A complex model becomes cumbersome to handle, more empirical in nature and difficult to ascribe physical reasons for the dependence of the response on the factors of the experiment.
- ❖ A noisy data fitted with too many parameters will fail to work well when implemented.

Adjusted R²:

Here Mean Squares of the error and model/total sum of squares are used.

$$R_{\text{adj}}^2 = 1 - \frac{\text{SS}_{\text{Error}} / n - p}{\text{SS}_{\text{Total}} / n - 1}$$

Adjusted R²:

$$R^2_{\text{adj}} = 1 - \frac{\text{SS}_{\text{Error}} / n - p}{\text{SS}_{\text{Total}} / n - 1}$$

To penalize the addition of unnecessary terms in the model equation (“overfitting”) the adjusted R² value also gets reported.

Adjusted R²:

$$R_{\text{adj}}^2 = 1 - \frac{\text{SS}_{\text{Error}} / n - p}{\text{SS}_{\text{Total}} / n - 1}$$

The SS_{Error} contribution is increased by dividing with the error degrees of freedom. The latter reduces when more number of parameters are taken up by the model.

Adjusted R²:

$$R_{\text{adj}}^2 = 1 - \frac{\text{SS}_{\text{Error}} / n - p}{\text{SS}_{\text{Total}} / n - 1}$$

Unless the SS_{Error} is considerably reduced by adding the extra term to the model equation, the adjusted R^2 will increase upon increasing the number of parameters.

PRESS: Prediction Error Sum of Squares

- ❖ This term is also similar to the sum of squares of the residuals.
- ❖ We sum the square of the deviations between the actual responses and the corresponding model predicted values.

PRESS: Prediction Error Sum of Squares

The main difference to watch out for is that the prediction for the i^{th} data is based on a model equation that *excluded* that particular data point but used the remaining data points to develop the model equation.

PRESS: Prediction Error Sum of Squares

- ❖ The same treatment is meted to other data points as well when they are compared to their individual model predictions.

PRESS:

First **H** matrix is defined as follows

$$\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$$

Let us define the usual residual

$$e_i = (y_i - \hat{y}_i)$$

The PRESS residual $e_{(i)}$ may be related to e_i as

$$e_{(i)} = e_i / (1 - h_{ii})$$

Here, h_{ii} refers to the diagonal elements of **H**.

PRESS:

The PRESS residual $e_{(i)}$ may be related to e_i as

$$e_{(i)} = \frac{e_i}{1 - h_{ii}}$$

Here, h_{ii} refers to the diagonal elements of **H**

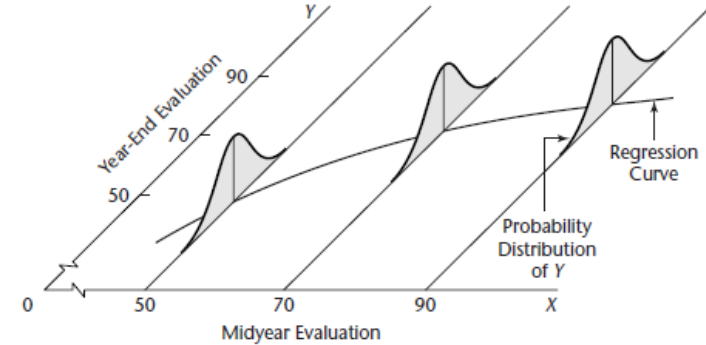
$$\text{PRESS} = \sum_{i=1}^n \left(\frac{e_i}{1 - h_{ii}} \right)^2$$

$$R_{\text{PRESS}}^2 = 1 - \frac{\text{PRESS}}{\text{TSSQ}}$$

Remember to adjust for intercept when finding total sum of squares

Representation of Regression

- ❖ Figure also shows probability distributions of Y for midyear evaluation levels $X = 50$ and $X = 70$
- ❖ Means of the probability distributions have a systematic relation to the level of X .
- ❖ This systematic relationship is called the **regression function of Y on X**
- ❖ Graph of the regression function is the regression curve
- ❖ The regression function is slightly curvilinear
- ❖ Increase in the expected (mean) year-end evaluation with an increase in midyear performance evaluation is retarded at higher levels of midyear performance.



Degrees of Freedom

$$SS_{\text{Total}} = SS_{\text{Regression}} + SS_{\text{Residual}}$$

The SS_{Total} has $n-1$ degrees of freedom

The $SS_{\text{Regression}}$ has k degrees of freedom where k is related to total number of regression coefficients as

Degrees of Freedom

$$p=k+1$$

The residual sum of squares has $(n-p)$ degrees of freedom where n is total number of responses.

Analysis of Variance (ANOVA)

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square
Regression	SS_R	k	MS_R
Error or Residual	SS_E	$n-p$	MS_E
Total	SS_T	$n-1$	

Source of Variation	Sum of Squares	Degrees of Freedom	Mean Square
Regression	SS_R	k	MS_R
Error or Residual	SS_E	$n-p$	MS_E
Total	SS_T	$n-1$	

Here n is the total number of observations and p is the total number of parameters (including the constant β_0).

$$\text{Since } k = (n-1) - (n-p)$$

$$\mathbf{k = p-1}$$